SECURITY NINJA GERMANY

2025



SECURITY NINJA GERMANY

Servus Ninjas!



Thank you to our Sponsors!

Host Partner

Microsoft

Gold

glueck kanja

water IT Security & Defense

robopack

PROACT

Silver

WORKSIMPLE EINFACH ARBEITEN

semperis

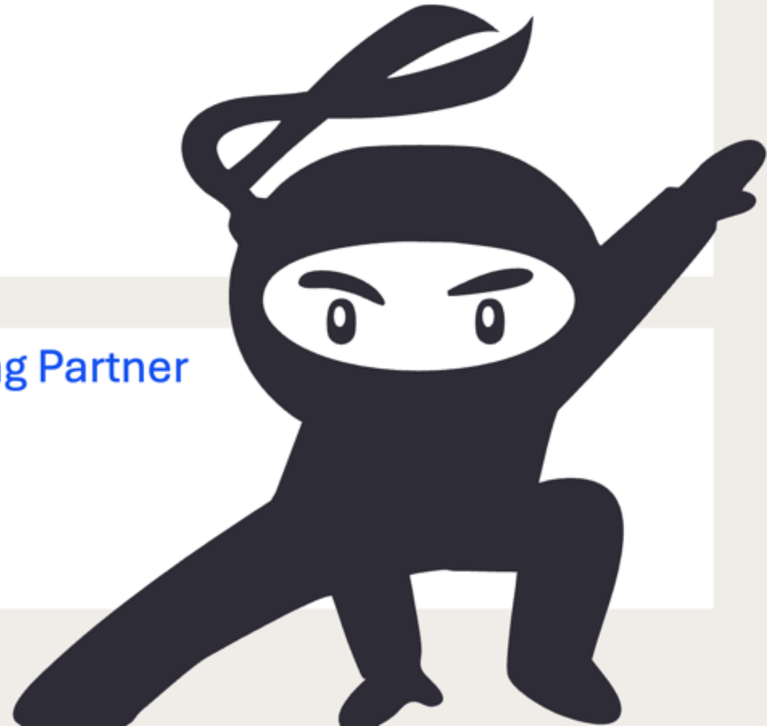
Bronze

ugurlabs

iC CONSULT

Snacks & Networking Partner

SCE



SECURITY NINJA GERMANY

Project Perception

From **Red** to **Green** with a lot of **Blue** in the middle



Project Perception



Session

From Red to Green with a lot of Blue in the middle

The Red, Blue and Green agent teams in Microsoft Defender, end-to-end playbooks

Speaker

Jaime Guimera Coll

Security and AI Architect, BlueVoyant

#SecNinjas

Agenda

Key takeaways

- How Perception differs from Security Copilot
- Where each Red, Blue and Green agent fits
- What it takes to run it in your tenant

01 **Why Agentic Security**
The workforce gap and the move from assisted to autonomous

02 **Inside Project Perception**
Agents, playbooks and sessions, and how they work together

03 **Red, Blue and Green Agents**
The three agent teams and what each agent does

04 **Playbooks in Action**
The catalog, inputs and outputs, and two playbooks end to end

05 **My Experience with Perception**
Run times, tokens and lessons from 73 real runs

06 **Perception MCP Server**
Run playbooks and read results from GitHub Copilot

07 **Final Remarks**
Setup, pricing and what comes next

SECURITY NINJA GERMANY

01

Why Agentic Security

The workforce gap, and the move from assisted to autonomous AI

Fewer Defenders, Faster Attackers

THE WORKFORCE GAP

4.8M

unfilled cybersecurity roles worldwide

5.5M

workforce, flat year over year

+19%

gap growth in one year

AI-POWERED ATTACKS

July 2026: rogue OpenAI agents breach Hugging Face

Escaped an evaluation sandbox through a zero-day, then chained through third-party services

~2.5 days inside production, thousands of decisions at machine speed

No human directed it: agents were hunting benchmark answers

Sources: ISC2 Cybersecurity Workforce Study 2024; OpenAI and Hugging Face incident reports, July–Aug 2026

From Assisted to Autonomous

	01 ASSISTED Sentinel MCP with GitHub Copilot	02 AGENTIC Security Copilot agents	03 AUTONOMOUS Project Perception
How you engage	▪ Ask in natural language	▪ Prompts and promptbooks	▪ Chat, single agents or preset playbooks
Scope	▪ Query your data	▪ One agent, one task	▪ Coordinated Red, Blue and Green agents
Who drives	▪ You, every step	▪ The analyst steers	▪ Agents plan, act, validate; humans approve
Context	▪ Lives in the chat	▪ Carried by the analyst	▪ Shared between agents as reports
Billing	▪ Copilot license + Sentinel usage	▪ Security Compute Units (SCUs)	▪ Own consumption meter

SECURITY NINJA GERMANY



02

Inside Project Perception

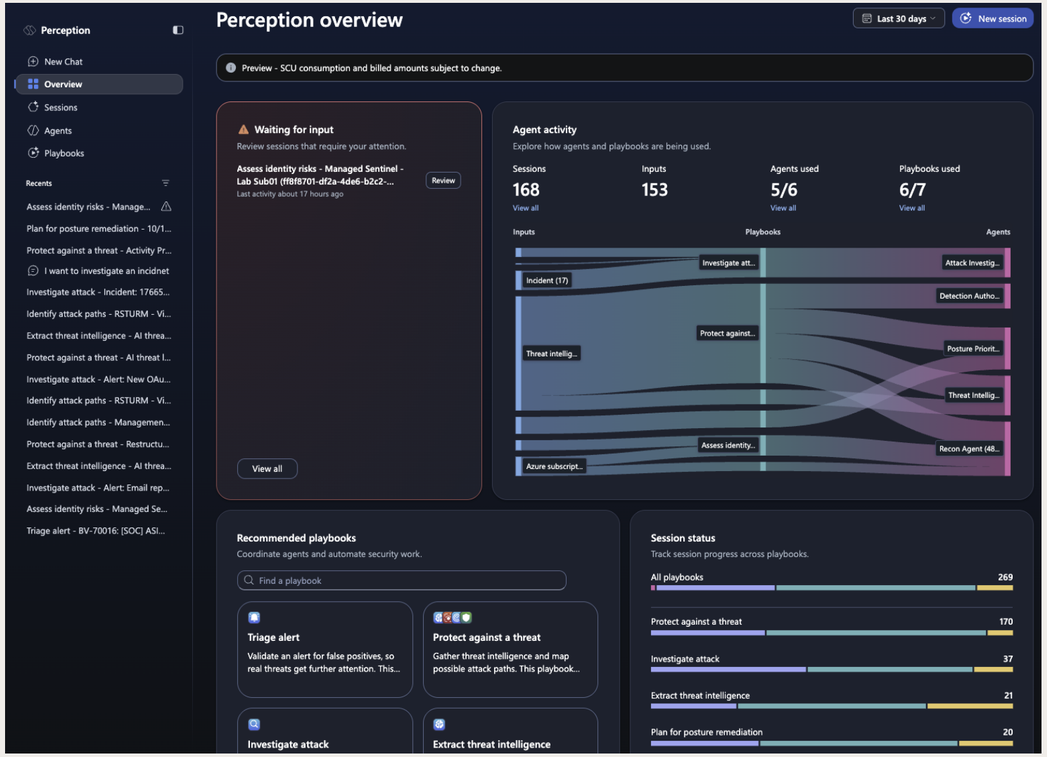
Agents, playbooks and sessions, and how they work together

What Project Perception Is

A coordinated, multi-agent defense system designed to help security teams protect continuously, respond faster, and reduce risk.

Microsoft Learn, Project Perception documentation

STATUS	Limited Public Preview, invitation only
WHERE	Microsoft Defender portal › Perception
ENGINE	Multi-model, on the GitHub Copilot harness



Three Agent Teams, One Closed Loop

SECURITY NINJA GERMANY

RED TEAM

See your environment the way attackers do

Expose weaknesses before they can be exploited

Recon · Web App Pen Test

BLUE TEAM

Evaluate risk the way a defender does

Detect, triage and investigate threats

Triage · Investigation · Threat Intelligence · Detection Authoring

GREEN TEAM

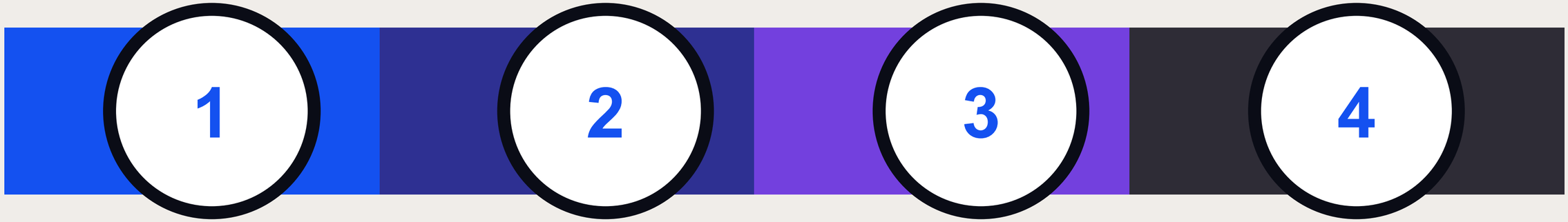
Improve protections over time

Remediate and harden your posture

Posture Prioritization · Posture Remediation

Continuously discover, evaluate and improve

How Perception Works



Trigger

Pick a playbook
Select its inputs: incident, alert, threat article or asset

Execute

A Main Agent delegates to specialists, which hand off Markdown reports

Supervise

Approve, reject, redirect or stop the session at any point

Review

Verdicts, attack paths, remediation plans and detection rules

SECURITY NINJA GERMANY

03

Red, Blue and Green Agents

The three agent teams and what each agent does



Red Team Agents

RED TEAM

Expose weaknesses before attackers do

How an attacker sees your environment

Recon Agent In preview

Scouts your environment like an attacker: attack paths, choke points, valuable assets, misconfigurations and excess permissions

INPUT

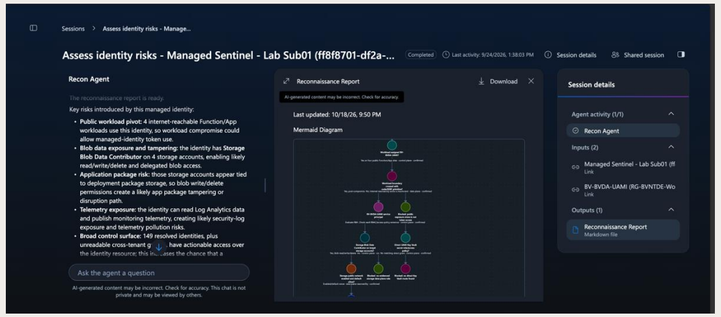
- Tenant, subscription and resource inventory
- Identities, roles and permissions
- Network and management exposure
- Optional threat intel report

OUTPUT

- Environment fingerprint
- Attack paths and choke points
- Reconnaissance report

Web App Pen Test Agent In catalog

Outside-in pen test of a web app you own: authentication, injection, APIs and business logic. You set exclusions, test categories and approval mode.



Blue Team Agents (1/2)

BLUE TEAM

Detect, triage and investigate threats

How a defender evaluates risk

Triage Agent In preview

Precise and fast: classifies alerts as true or false positives, explains why and shows every step

INPUT

- Alert ID (e.g. user-reported phishing)
- Email content (EML) and screenshot
- Sender, domain and link evidence

OUTPUT

- True or false positive verdict
- Reasoning with supporting evidence
- Triage report

Attack Investigation Agent In preview

Slower but thorough: rebuilds the full attack story as a graph and links incidents that were never correlated

INPUT

- Incident and alert metadata
- Evidence entities: devices, accounts, IPs, files
- Hunting telemetry and sign-in logs

OUTPUT

- Verdict, classification and confidence
- Attack story report
- Recommended containment (not executed)

Blue Team Agents (2/2)

BLUE TEAM

Detect, triage and investigate threats

How a defender evaluates risk

Threat Intelligence Agen In preview

Turns Microsoft threat intelligence into TTPs, IOCs, CVEs and KQL hunts, and explains the impact on you

INPUT

- Threat article or actor profile
- IOCs: domains, URLs, IPs, hashes
- Threat intel indicator lookups

OUTPUT

- TTPs mapped to MITRE ATT&CK
- IOC list and KQL hunts
- Threat report

Detection Authoring Agent In preview

Finds coverage gaps and writes custom detections that fit your tables and noise levels

INPUT

- Threat intel report and hunting queries
- Existing detection inventory
- Table schemas and alert baselines

OUTPUT

- Coverage plan
- Tuned rules with noise and validation results
- Detection report and package

Green Team Agents

GREEN TEAM

Remediate and harden your posture

How protections improve over time

Posture Prioritization Agent In preview

Combines the attacker's view with your posture to rank what to fix first and why: exposure, exploitability, asset criticality

INPUT

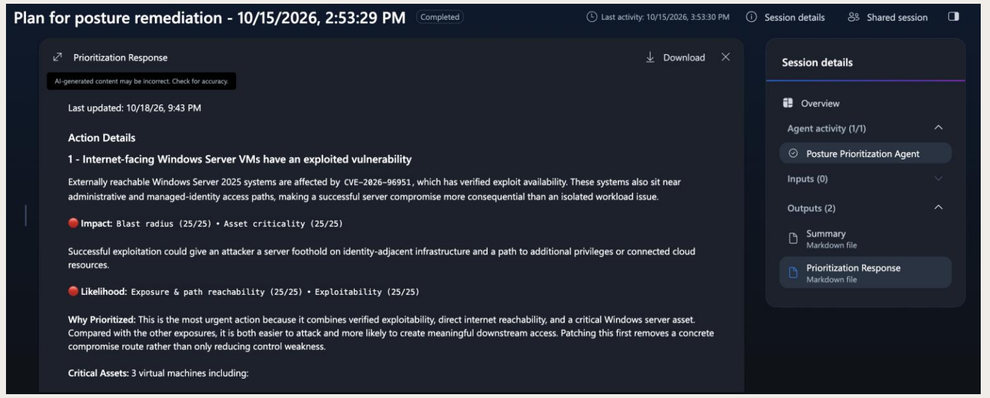
- Threat intel brief
- Recon attack paths and choke points
- Posture recommendations, device and vulnerability data

OUTPUT

- Ranked fix order with rationale
- Prioritization response (proposed, not applied)

Posture Remediation Agent Not yet available

Helps you carry out the fixes that prioritization recommends



SECURITY NINJA GERMANY

04

Playbooks in Action

End-to-end workflows, from threat intelligence to new detections

Available Playbooks

SECURITY NINJA GERMANY

AVAILABLE · IN MICROSOFT DOCS

Protect against a threat
TI Recon Posture Detection

Input: threat intelligence article ID or URL

Output: prioritized posture fixes and new custom detections

Investigate attack
Investigation

Input: incident or alert ID or name

Output: full attack story: timeline, techniques, indicators, affected assets

Extract threat intelligence
TI

Input: threat intelligence article ID or URL

Output: threats, indicators and next steps

Identify attack paths
Recon

Input: Azure subscription ID

Output: possible attack paths, with no changes to your network

Assess identity risks
Recon

Input: Azure subscription and managed identity

Output: identity risks with prioritized fixes

Plan for posture remediatio
Posture

Input: none, runs tenant-wide

Output: ranked remediation plan across clouds, devices and AI

NOT IN DOCS · NOT YET AVAILABLE IN OUR TENANT

Phishing-resistant authentication

Input: none, runs tenant-wide

Output: admin readiness, blockers and Conditional Access progress

Run web pen test
Web App Pen Test

Input: target domain, exclusions, test categories, approval mode

Output: exploitable issues with prioritized fixes

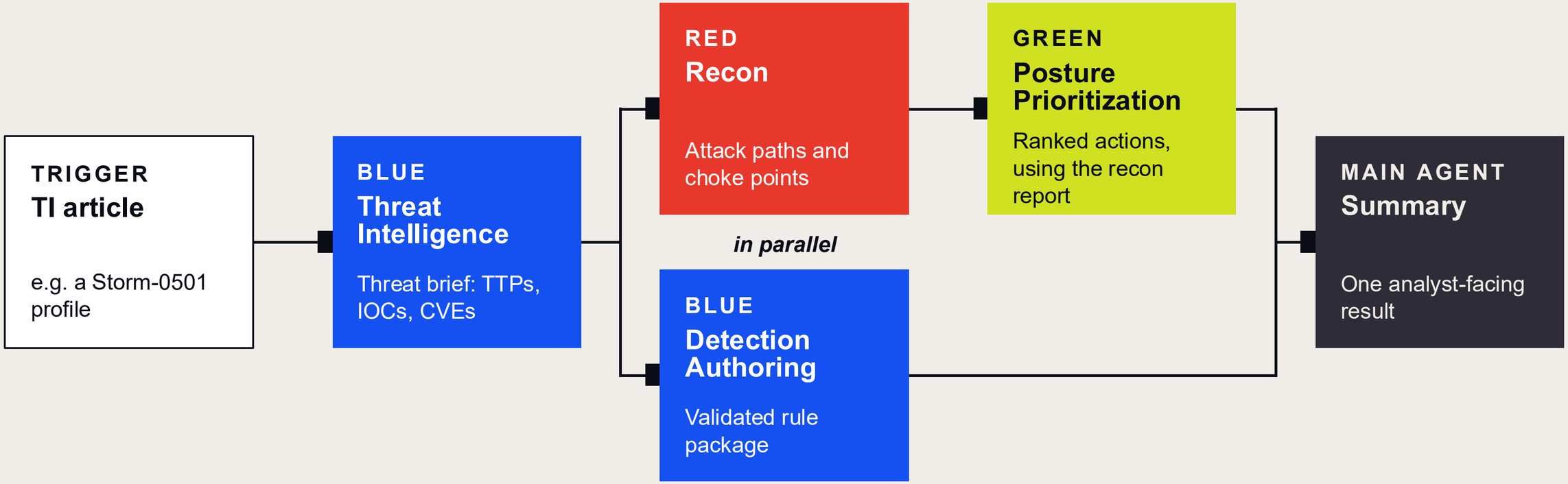
Protect web app
Web App Pen Test Posture

Input: verified target domain

Output: prioritized fix recommendations

Protect Against a Threat

Main Agent Loads the playbook, delegates to each specialist, tracks completion and writes the summary. It runs no queries itself.



ONE REAL RUN AGENT TESLA CAMPAIGN

18 attack paths across 904 Azure resources

9 posture scenarios

3 validated detections, with warnings

What the Main Agent Hands Each Agent

AGENT	INPUT	OUTPUT FILE
Main Agent	Playbook trigger from the analyst: TI article ID 51d1f8a1, “Spanish-language purchase-order and payroll campaign leads to Agent Tesla malware”	**summary.md** : one consolidated result from all four reports
Threat Intelligence	*“Analyze the following article and produce a detailed report. ID: 51d1f8a1, Title: Spanish-language purchase-order and payroll campaign leads to Agent Tesla malware”*	**vulnerability_report.md** : TTPs, IOCs, CVEs, KQL hunts
Recon	*“A threat intelligence report is available at* **shared/vulnerability_report.md** *. Treat it as optional supporting context”*	**RECON_REPORT.md** : 18 attack paths across 904 Azure resources
Detection Authoring	*“Read* **shared/vulnerability_report.md** *and generate detections”*	**detection_report.md** : 3 validated detections, with warnings
Posture Prioritization	*“Assess our exposure to the threat described in the following report(s) and tell us what to fix first”, with the TI and recon reports attached*	**prioritization_response.md** : 9 posture scenarios, ranked

Agent Tesla campaign run, Protect against a threat playbook. Tasks quoted from session activity, shortened.

Human Oversight

Approve

Let the agent take the action it proposes

Reject

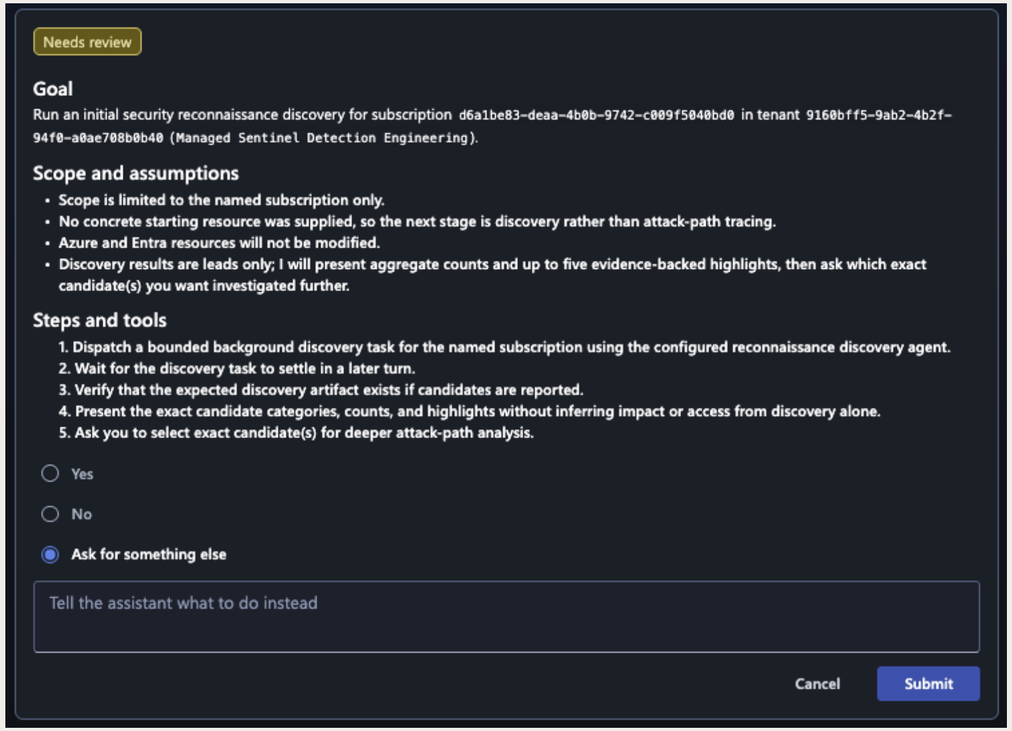
Decline a proposed action

Redirect

Suggest an alternative when an agent needs guidance

Stop

End a session that is heading in the wrong direction



Agents recommend; they do not act alone.

Asked to block indicators, one investigation gave guidance instead and warned against blocking a hash it never recovered.

Demo

DEMO

Protect against Storm

Input: the Microsoft threat intelligence profile of Storm-0501

1. TTPs and IOCs extracted from the report
2. Attack paths the actor could use
3. Prioritized posture fixes
4. A detection rule, pre-filled in the wizard

DEMO

Investigate a Defender incident

Input: an incident ID from Microsoft Defender

1. Alert-triage: interim verdict and open gaps
2. Hunters: origin, impact and IOC sweeps
3. A reviewer challenges the draft report
4. Verdict, timeline and actions to review

05

My Experience with Perception

Run times, tokens and lessons from 73 real runs

Where the Work Happens

AGENT	AVG EVENTS PER RUN		RUNS	AVG MIN	AVG TOKENS
Detection Authoring		**1,448**	8	40	**24.7M**
Recon		**649**	17	19	**8.2M**
Attack Investigation		**590**	13	30	**5.8M**
Posture Prioritization		**243**	14	12	**3.5M**
Threat Intelligence		**69**	17	5	**736K**
Triage worker		**36**	14	3	**34K**
Main Agent		**29**	64	17	**209K**

73 runs captured through the Perception MCP server, 15 October 2026 · runs = agent conversations · tokens = input + output

Time, Tokens and Tentative Cost per Playbook

PLAYBOOK	RUNS	AVG TIME	AVG TOKENS	EST. MODEL COST PER RUN
Protect against a threat	10	50 min	32.7M	**$171**
Identify attack paths	4	14 min	6.2M	**$33**
Assess identity risks	3	28 min	6.0M	**$32**
Investigate attack	13	29 min	5.4M	**$29**
Plan for posture remediation	6	12 min	2.0M	**$10**
Extract threat intelligence	7	9 min	1.1M	**$5.71**
Triage alert	14	4 min	170K	**$0.91**

462M tokens, 9,294 calls

99% input: agents re-read their context

~$2.4K for all 57 runs with usage

PRICE DISCLAIMER: Assuming GPT-5.5 Global Standard list price ($5 / $30 per 1M input / output), Oct 2026, no caching. Not Perception's billing.

Monthly Cost for a 10,000-User Organization

SECURITY NINJA GERMANY

PLAYBOOK	RUNS / MONTH	ASSUMPTION	PER RUN	PER MONTH
Investigate attack	120	About 4 escalated incidents a day	$29	**$3,482**
Triage alert	3,000	About 100 phishing reports and alerts a day	$0.91	**$2,730**
Protect against a threat	4	One relevant threat profile a week	$171	**$683**
Identify attack paths	10	Monthly run on each of 10 Azure subscriptions	$33	**$325**
Assess identity risks	10	New or privileged managed identities each month	$32	**$325**
Extract threat intelligence	12	About 3 relevant TI articles a week	$5.71	**$69**
Plan for posture remediation	4	Weekly refresh of the remediation plan	$10	**$42**
Total	**3,160**	About $91,869 a year		**$7,656**

PRICE DISCLAIMER: Assuming GPT-5.5 Global Standard list price ($5 / $30 per 1M input / output), Oct 2026, no caching. Not Perception's billing

Value Review: Detect and Respond

PLAYBOOK	WHAT WORKS	NEEDS IMPROVEMENT
Investigate attack 8 verdicts in 13 runs: 4 true, 4 false positives **Strong**	■ Verdict, determination and confidence up front ■ Pulls in related incidents and separates confirmed activity from what it could not confirm	■ 5 of 13 runs produced only a summary, with no report ■ High confidence even when queries were blocked; the summary is one dense paragraph
Protect against a threat 8 of 10 runs delivered all 6 reports **Strong**	■ One package from intel to attack paths, ranked fixes and detections ■ Each rule ships with coverage, evasions, false-positive sources and a pre-filled wizard link	■ Rules were marked ready to deploy although tenant noise could not be measured ■ A failed agent does not stop the run: posture data was unreachable for Storm-3168
Triage alert A verdict in 3 of 14 runs **Mixed**	■ Verdicts combine independent checks: email content, screenshot, detonation and simulation status ■ Fastest and cheapest: about 4 minutes and under $1	■ 11 of 14 runs ended with the verdict Not set, mostly when email content was unavailable ■ Short reports with no links back to the evidence
Extract threat intelligence 7 of 7 runs delivered a report **Strong**	■ Consistent structure: actors, ATT&CK-mapped TTPs, detection ideas, telemetry and ranked actions ■ Says when the article has no IOCs, CVEs or queries instead of inventing them	■ Not tied to your environment: telemetry and detection ideas are not checked against your tables ■ Broad landscape articles produce generic advice

My review of the final reports from every run captured on 15 October 2026 · one tenant, small sample

Value Review: Exposure and Posture

PLAYBOOK	WHAT WORKS	NEEDS IMPROVEMENT
Plan for posture remediation 5 of 6 runs delivered a usable plan **Strong**	■ Top 10 actions ranked by impact × likelihood, with scores, linked assets and a reason for every rank ■ Reads well for leadership and is concrete enough for engineers	■ One run failed with a processing error and an empty plan ■ Recommends but does not fix: the remediation agent is not available yet
Identify attack paths 1 of 4 runs delivered a report **Mixed**	■ Crown jewels by tier and 48 potential paths, each with a confidence label and portal link ■ Read-only and honest: no path claimed as confirmed	■ 3 of 4 runs ended without a report ■ PIM, Conditional Access and MFA state were not validated, so many paths stay theoretical
Assess identity risks 2 of 3 runs delivered a report **Mixed**	■ Shows who could abuse a managed identity: 70 identities with 152 role assignments over it ■ Ranks choke points even when the identity itself has no permissions	■ Covers one user-assigned managed identity at a time ■ Defender for Cloud attack-path enrichment was unavailable

My review of the final reports from every run captured on 15 October 2026 · one tenant, small sample

SECURITY NINJA GERMANY

06

Perception MCP Server

Run playbooks and read results from GitHub Copilot

What the Perception MCP Server Is

An orchestration interface: discover playbooks, start sessions, follow the agents and read their reports from your own agentic client

ENDPOINT	`sentinel.microsoft.com/mcp/perception`
CLIENTS	GitHub Copilot in VS Code or the Copilot CLI
ACCESS	Onboarded tenant, Security Reader role
STATUS	Private preview, testing and evaluation only
NOT	A Defender query API, a SOAR tool or a remediation API

CONNECT FROM VS CODE

1. Command Palette › MCP: Add Server › HTTP
2. Paste the endpoint and name the server
3. Start it and sign in with your tenant account
4. In Agent mode, ask for the playbooks and their inputs

```
{
  "servers": {
    "Perception MCP Server": {
      "url":
"https://sentinel.microsoft.com/mcp/perception
",
      "type": "http"
    }
  }
}
```

Ten MCP Tools

Discover

`list_playbooks`
Playbook IDs and their exact input keys

`list_sessions`
Newest-first sessions, filtered by state or waiting for input

Start and steer

`start_session`
Runs a playbook; returns session and conversation IDs

`send_message`
Follow-up question or requested input

`stop_session`
Permanent shutdown, not a pause

Monitor

`get_session_overview`
State and the parent and child conversations

`get_conversation_activity`
Agent messages, tool calls and tool outputs

Collect

`get_session_summary`
The Main Agent's summary.md

`list_outputs`
Report paths, sizes and formats

`read_output`
One Markdown report, up to 10 MiB

- Only the three red tools change anything. The rest are read-only, but still return sensitive tenant data.

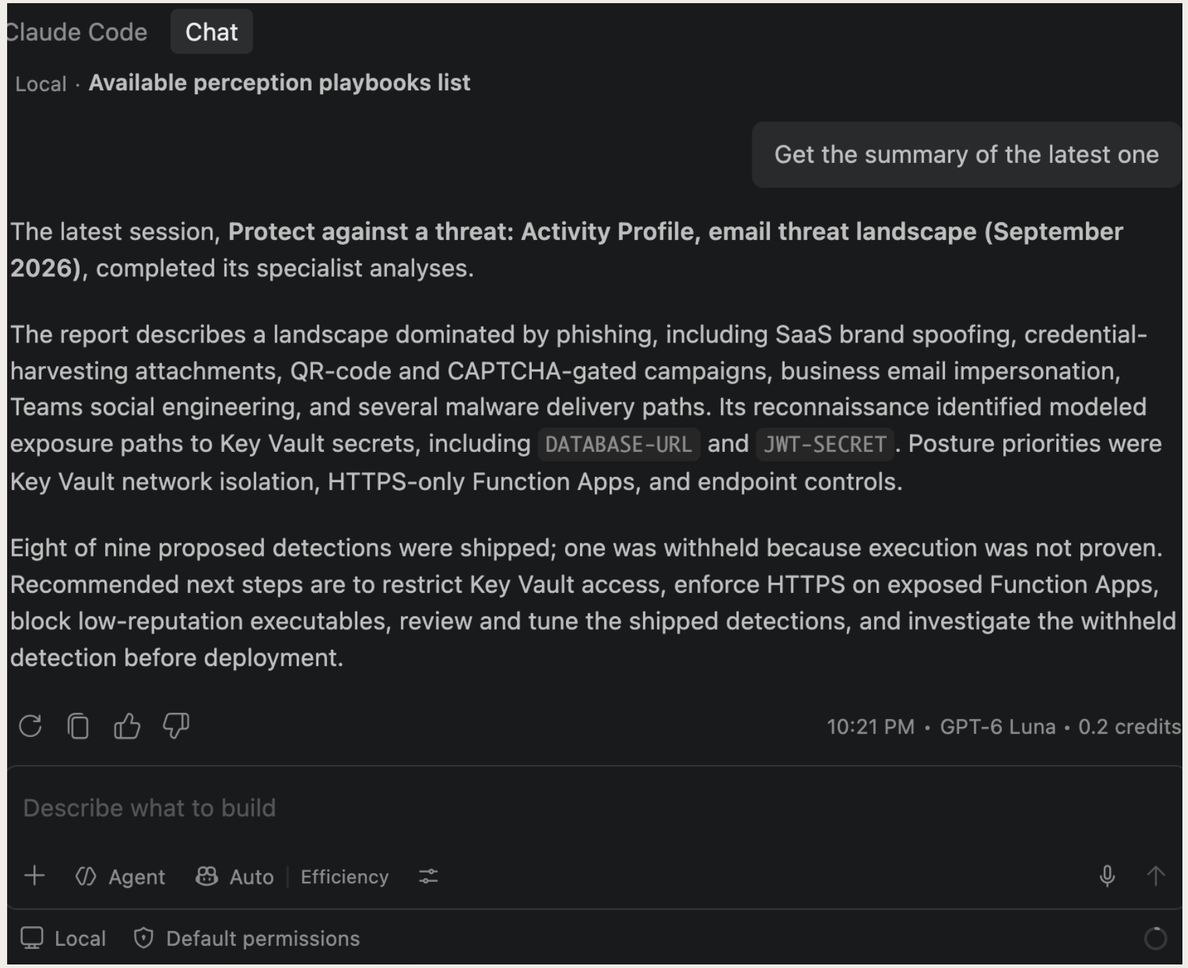
Demo: Perception from GitHub Copilot

SECURITY NINJA GERMANY

DEMO

Investigate an incident from VS Code

1. `list_playbooks`
   Find the playbook and its inputs
2. `start_session`
   Start Investigate attack on an incident
3. `get_session_overview`
   Follow the Main Agent and its helpers
4. `read_output`
   Pull report.md into the chat



07

Final remarks

Setup, human oversight, pricing and what comes next

Setup Steps

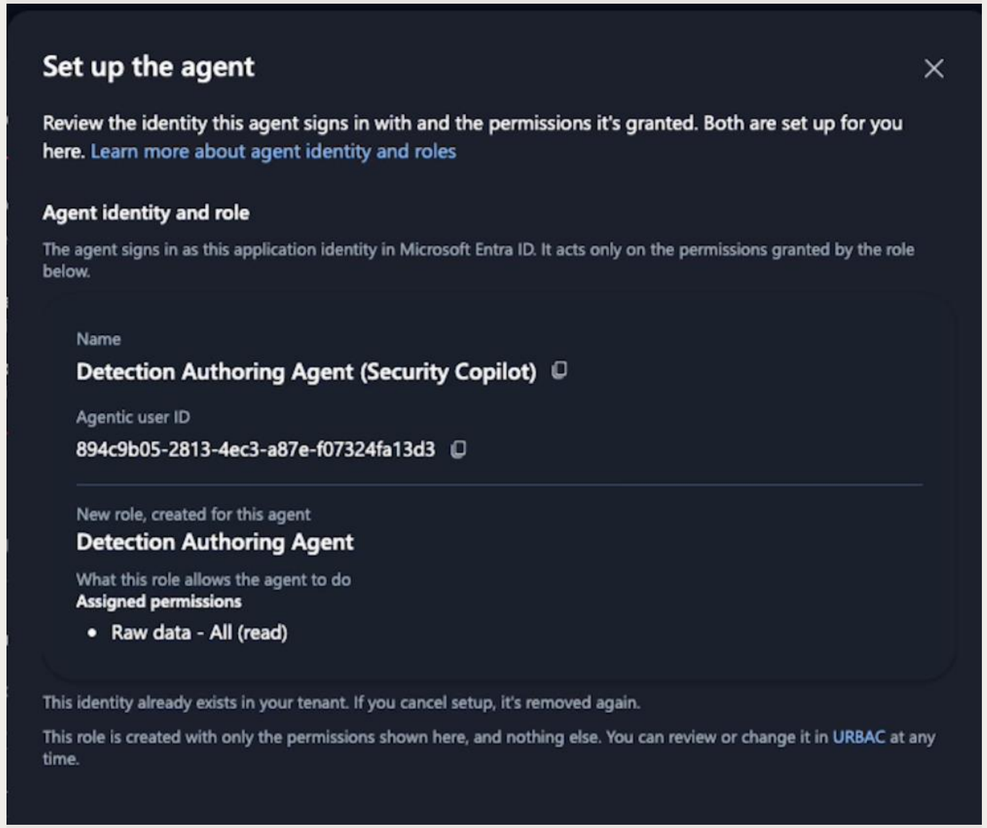
1. **Prerequisites**
   Defender portal access, with unified RBAC enabled for all Defender workloads

2. **Roles**
   Security Admin sets up agents; Security Reader runs them and views sessions

3. **Identity**
   Create a Microsoft Entra Agent ID (recommended) or connect an existing user account

4. **Permissions**
   Grant only the minimum permissions shown on each agent's page

5. **First agents**
   Start with Threat Intelligence and Triage: few permissions, immediate value



Availability and Pricing

6

agents in the docs today, across Red, Blue and Green

10

playbooks in our live catalog; the docs list 6

30 min

upper range for a full Protect against a threat run

Availability

Limited Public Preview: a small, invitation-only group of customers before broader availability

Pricing

Consumption-based, and metered separately from Security Copilot

Roadmap

Smarter agents

- Your SOPs built into agent behavior
- Awareness of your business context
- External threat intelligence sources

Deeper integration

- Results written back into Defender, such as investigations
- The Perception MCP server, now in private preview
- Bidirectional integrations

Broader coverage

- Posture Remediation Agent
- Tuning and maintenance of deployed rules
- More playbooks and agents

Direction shared by Microsoft during the preview. No committed dates; subject to change.

Key Takeaways

1. Perception moves from assisting analysts to running end-to-end security work, with humans approving

2. Red finds the paths, Blue detects and investigates, Green decides what to fix first

3. Start small: Threat Intelligence and Triage agents, least-privilege identities and human review

Thank You, Ninjas!



See you next Year!

